SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Yazhini·R (192521190) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R·Yazhini

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

...uestions

3. An e-learning platform experiences data loss due to accidental deletion in the cl... A15
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.
   Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
   Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

ompany collects terabytes of customer transaction data every day from online and offline
CO The management wants to analyze purchasing patterns in near real time to improve product
ecommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks.
- Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

- Design a secure cloud architecture that protects sensitive healthcare data.
- Explain how encryption, identity and access management (IAM), and key management should be implemented.
- Discuss auditing, logging, and compliance monitoring strategies.
- Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

- Design a serverless cloud architecture for the application.
- Explain how event-driven computing can improve scalability.
- Discuss the role of managed databases, API gateways, and message queues.
- Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

- Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources.
- Explain how data quality, metadata management, and governance should be maintained.
- Discuss the role of workflow orchestration and scheduling tools.
- Recommend strategies for optimizing query performance and storage costs.

5.A logistics company wants to predict delivery delays using historical shipment information, and live GPS tracking. The analytics platform must support continuous real-time predictions.

- Design a cloud-based machine learning and analytics architecture.
- Explain how big data technologies can support model training and inference.
- Discuss the use of cloud-native AI/ML services for model deployment and monitoring.
- Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.

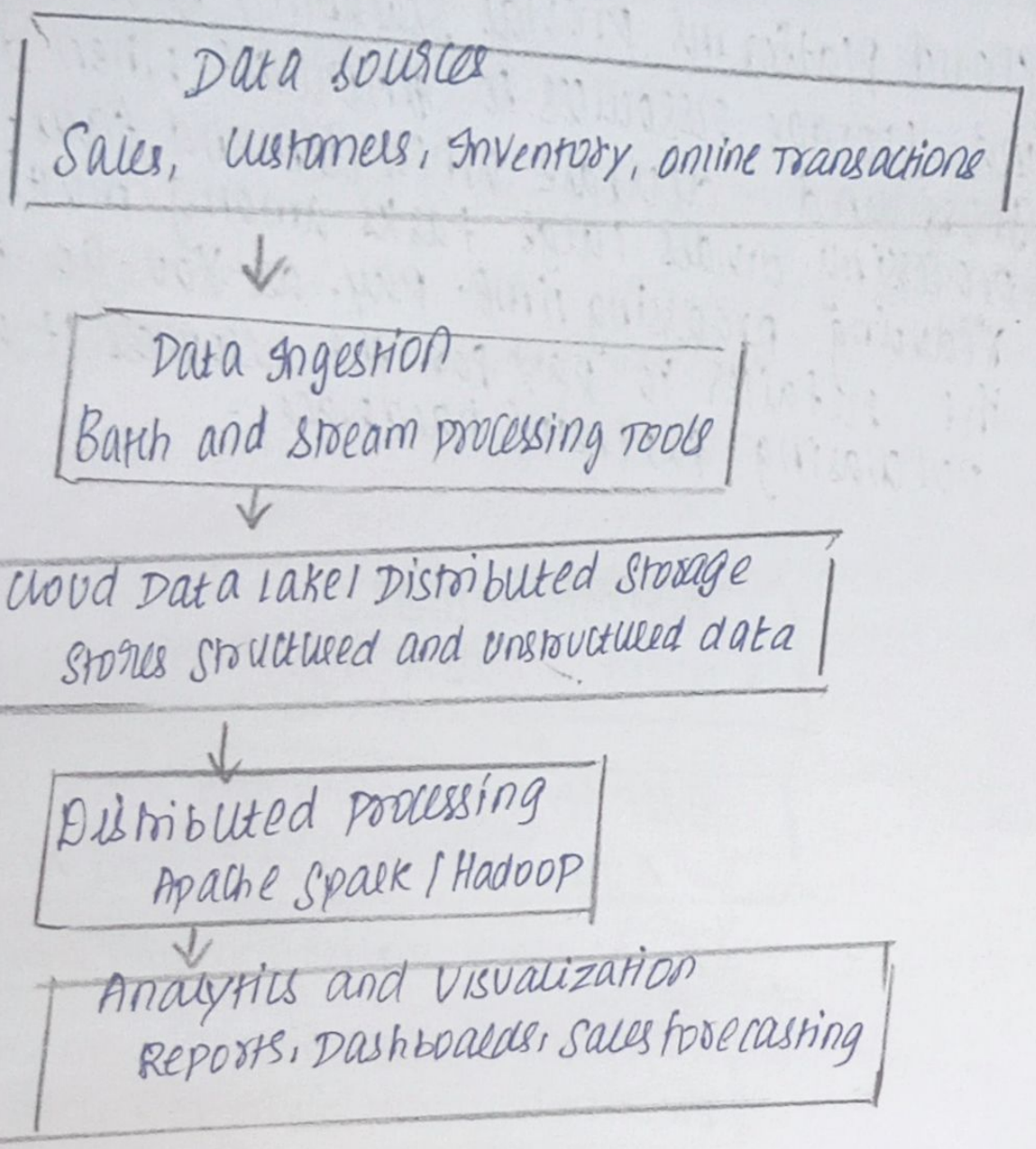
...oud monitoring tools, metrics, logs, and tracing help identify performance ...ks.

- ...scuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

a)



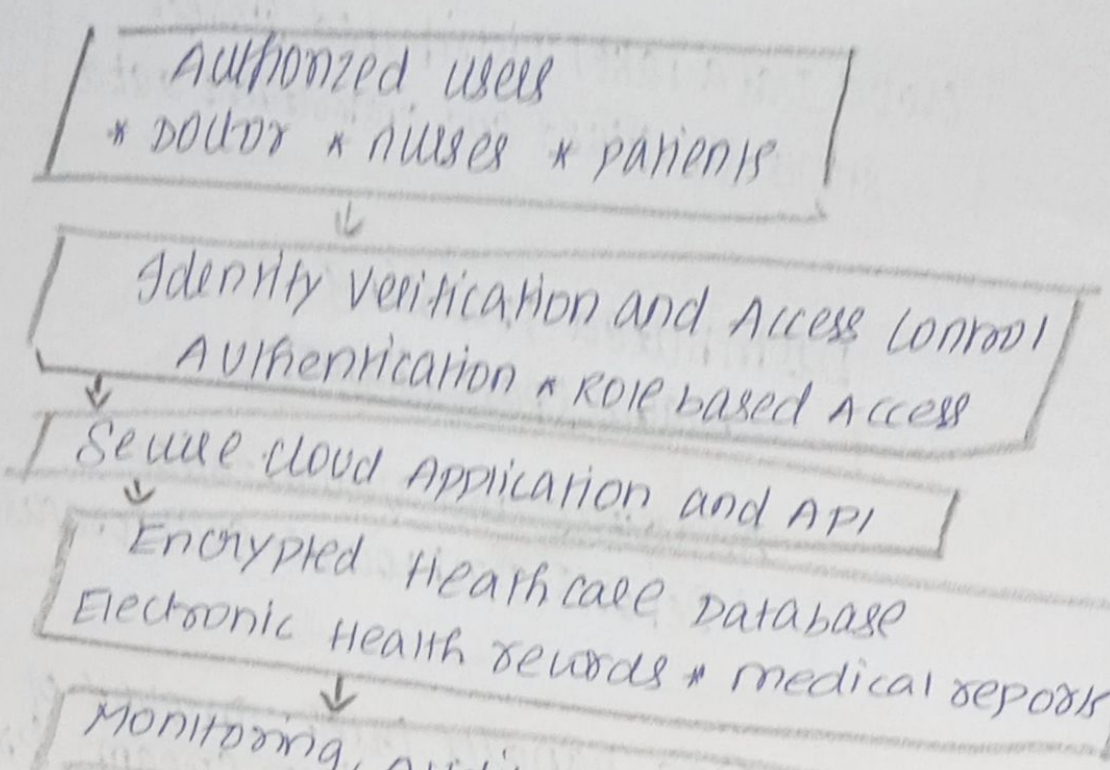
b) Batch processing handles large amounts of stored data at scheduled intervals, while stream processing analyzes data continuously as it arrives. A retail company can use batch processing to calculate monthly sales and stream processing to monitor live purchases. Integrating both methods provides historical analysis and real-time insights, helping the company respond quickly to customer demand.

c) A data lake stores large amount of raw data in different formats, including text, images, and transaction records. A data warehouse stores cleaned and structured data for reporting and analysis. The retail company can first store all collected data in a data lake and then transfer relevant, processed information into a data warehouse

d) Cloud platforms provide scalability by allowing compute and storage resources to increase when data volumes grow and decrease when demand falls. Distributed processing divides large tasks among multiple machines, reducing processing time. Pay-as-you-go pricing allows the retailer to pay for the resources it uses instead of purchasing expensive hardware.

2.

a)



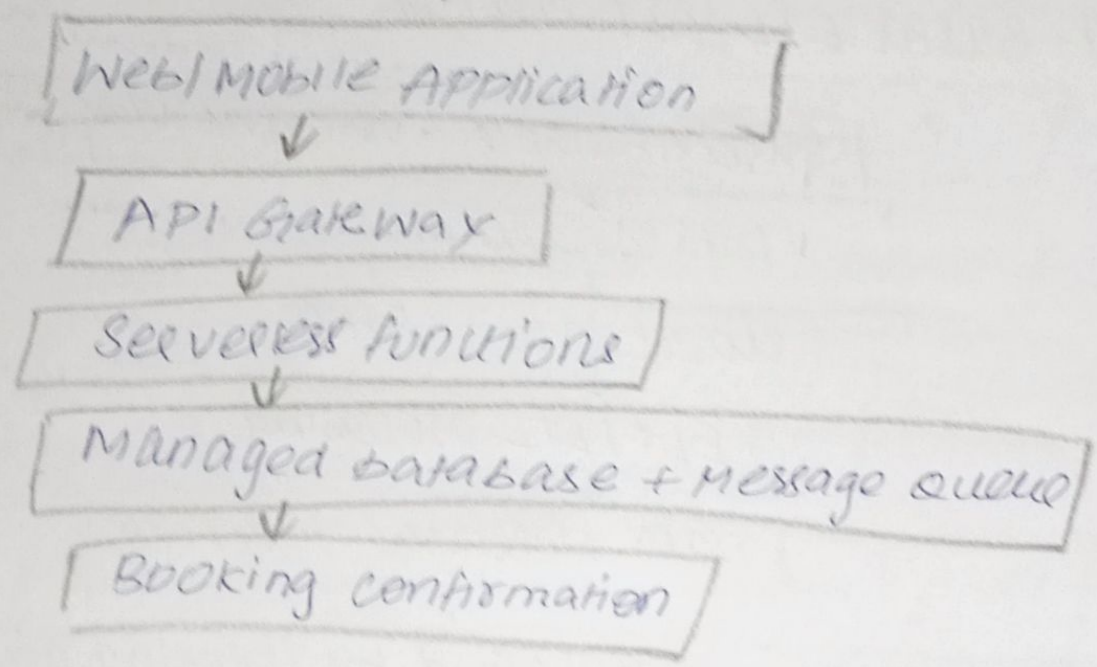
b) Encryption converts readable medical information into an unreadable form that can only be accessed using the appropriate decryption key. Data should be Encrypted both during transmission and while stored Identity and Access Management (IAM) verifies users and assigns permissions according to their roles. key management securely stores, creates, rotates and controls Encryption keys. Together, these mechanisms protect sensitive healthcare records from unauthorized access

...ring and logging record user logins, medical record access, and data changes continuous monitoring detects suspicious activities and generates alerts. Regular audits and compliance checks ensure that healthcare data follows privacy and security regulations

d) use multi-factor authentication, strong pass words, role-based access control, and encryption to protect patient data. Firewalls and regular security updates prevent attacks. Backups, employee training, and continuous monitoring help reduce data breaches.

8) Serverless ticket Booking system.

a)



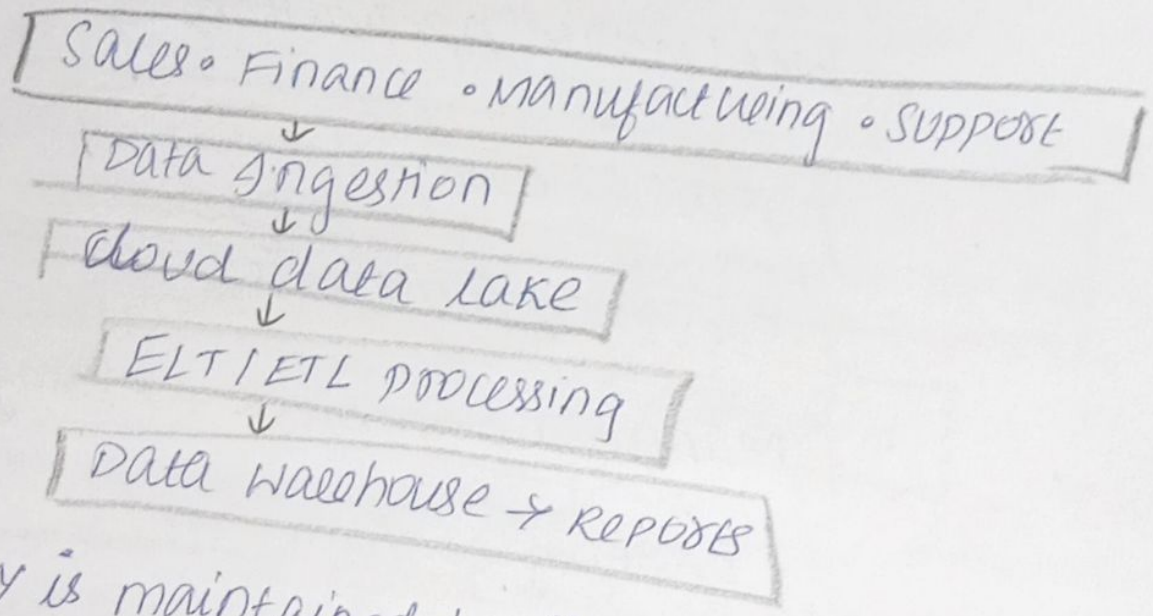
b) Event-driven computing executes functions when events such as booking requests or cancellation occur. During traffic spikes, the cloud platform automatically increases function instances to handle more requests when traffic decreases, resources reduce, improving response time and cost efficiency

c) API gateways manage incoming requests and the correct functions. Managed databases store customer and booking information reliably. Message queues handle background tasks such as sending confirmation Emails, reducing workload on the main application

d) Serverless computing offers automatic scaling, reduced sever maintenance, and pay-per-use pricing. However, cold starts, Execution limits, and cloud provider dependency can be limitations. Proper transaction handling is essential to prevent duplicate bookings and multiple customers booking the same seat

4. Cloud-Based ETL/ELT Pipeline

a)



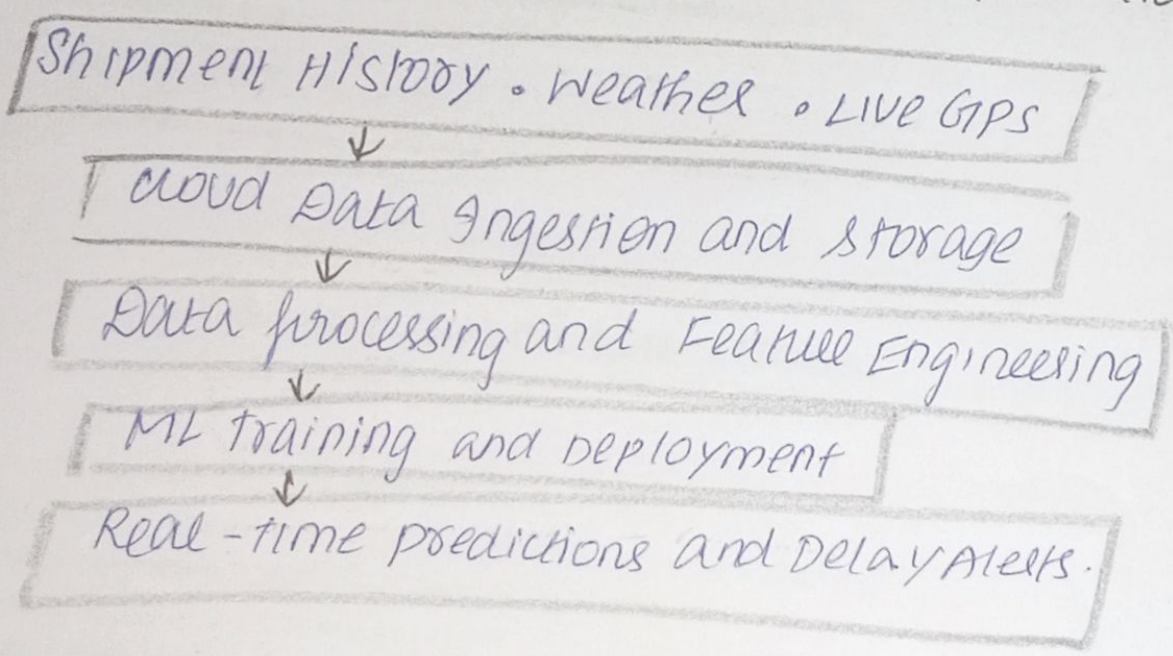
b) Data quality is maintained by checking accuracy, completeness, consistency, and duplicates. Metadata records the data's source, format, and meaning. Data governance defines ownership, access permissions, privacy policies and security rules to ensure reliable and controlled data usage

→ Flow orchestration coordinates pipeline tasks, including data Extraction, transformation, validation intervals or when new data arrives. Automatic retries failures and reduce manual work.

d) Use partitioning, compression, suitable file formats, and Indexing where supported to improve query performance. Remove unnecessary data and optimize queries to reduce processing time. Use low-cost storage for rarely accessed data and monitor cloud resource usage to control expenses.

5. Cloud-Based learning - for Delivery Delay prediction

a)



b) Big data Technologies such as Apache Spark process large shipment datasets efficiently across multiple machines. Historical data is used to train models and identify delay patterns. During inference, current GPS, traffic, and weather data are processed to generate fast delivery predictions.

c) Cloud AI/ML services provide tools for model training, deployment, and monitoring. Models can be deployed as APIs to provide real-time predictions. Monitoring tracks accuracy, response time, and data changes, while alerts help identify problems and determine when retraining is needed.

d) Use streaming tools for live GPS data and batch pipelines for historical records. Feature engineering creates useful variables such as distance, speed, weather, and previous delays. Regularly retrain models with new delivery outcomes and monitor performance to maintain prediction accuracy.